

Flask

1. Understand the basics of Flask

→ Install Flask:

```
pip install flask
```

→ Basic "Hello, world" Example

```
from flask import flask
```

```
app = flask(__name__)
```

```
@app.route('/')  
def home():
```

```
    return "Hello, world"
```

```
if __name__ == "__main__":
```

```
    app.run(debug=True)
```

Flask is initialized using `flask(__name__)`

Routes defines what URL's trigger specific functions

`app.run(debug=True)` enables debugs during development

to run

load a model using flask

```
import pickle  
from flask import request
```

```
# Load the model
```

```
with open ('model.pkl', 'rb') as file:  
    model = pickle.load(file)
```

```
@app.route ('/predict', methods=['post'])  
def predict():  
    data = request.json  
    prediction = model.predict([data['features']])  
    return {'prediction': prediction.tolist()}
```

Add Error handling

```
@app.errorhandler(404)  
def page_not_found(e):
```

```
    return {"error": "page not found"}, 404
```


3. Run the Application

```
python app.py
```

→ Using HTML Templates

```
index.html
```

```
<!DOCTYPE html>
```

```
<html>
```

```
  <title> Home </title>
```

```
</head>
```

```
<body>
```

```
  <h1> welcome to flask </h1>
```

```
</body>
```

```
</html>
```

app.py now

```
from flask import render_template
```

```
@app.route('/')
```

```
def home():
```

```
    return render_template('index.html')
```